

Enhanced BCI Performance using Diffusion Model for EEG Generation

Yucun Zhong, Lin Yao*, Yueming Wang

Abstract—In the realm of Motor Imagery (MI)-based Brain-Computer Interface (BCI), the widespread adoption of deep learning-based algorithms has resulted in an increased demand for a larger training sample size, thereby placing a heightened burden on users. This study advocates the utilization of one of the most advanced generative models, the denoising diffusion probabilistic model (DDPM), for the artificial synthesis of Electroencephalogram (EEG) raw signals. The quality of the generated EEG signals is evaluated through both qualitative and quantitative analyses. Through dimensionality reduction projection, we observed a notable similarity in the data distributions between the generated EEG signals and real EEG signals. Additionally, spectral analysis indicates a striking similarity in energy distribution between the two, accompanied by the presence of an event-related synchronization (ERS) phenomenon in the generated EEG signals. Quantitative analysis reveals that the accuracy of generated EEG signals for left and right-hand motor imagery tasks is $89.81 \pm 2.11\%$, with discriminative information related to classes predominantly concentrated in the motor-sensory cortex area and alpha-beta frequency band. Furthermore, the integration of generated EEG samples contributes to a 3.17% improvement in the classification performance of BCI-deficiency subjects. These artificially generated EEG signals exhibit promising potential for application in calibrating MI-BCI deep learning models, thereby alleviating the burden on participants.

Index Terms—Brain-computer interface, motor imagery, denoising diffusion probabilistic model (DDPM), BCI-deficiency

I. INTRODUCTION

Since the advent of Motor Imagery (MI)-based Brain-Computer Interfaces (BCI) [1], a recurrent challenge has come to light—roughly 30% or even more of the subject population encounters a binary recognition rate below the acceptable threshold of 70% for the two-handed motor imagery task [2]. This challenge is commonly known as the “BCI-deficiency” problem [3]. Recent research endeavors have offered promising solutions to mitigate this issue through the application of deep learning [4]. However, the calibration process for deep learning models typically necessitates a considerable volume of data. If this data is directly collected from users, it poses a substantial burden, ultimately adversely affecting the overall user experience.

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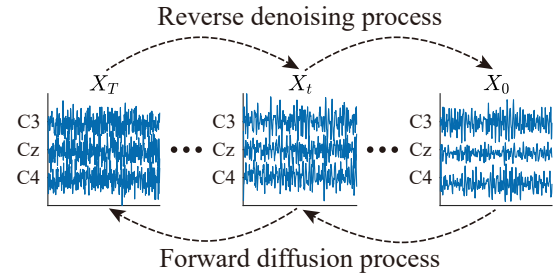


Fig. 1. The illustration of DDPM in the context of EEG generation.

Addressing the challenge of insufficient sample size involves the utilization of generative models for sample generation [5]. At the forefront of generative models is the Denoising Diffusion Probabilistic Model (DDPM) [6], renowned for its clear definition, efficient training, and controllable generation capabilities, particularly in image generation [6]. While DDPM has demonstrated its dominance across various domains, the generation of EEG signals has predominantly relied on Generative Adversarial Networks (GAN) [7], [8]. However, in contrast to DDPM, GAN-based methods pose challenges in terms of training complexity and definition clarity. Additionally, although algorithms similar to DDPM have been initially employed in EEG generation, their primary focus has centered on generating covariance matrices rather than raw EEG signals [9]. This limitation restricts their application to classification methods utilizing covariance as input. In our current research, we opt for DDPM in the generation of EEG raw signals instead of relying on intermediate features such as covariance or neural network feature vectors. This strategic choice aims to capitalize on the inherent strengths of DDPM in capturing the intricate complexities and nuances present in raw EEG data.

As depicted in Figure 1, DDPM is composed of two processes: a forward diffusion process and a reverse denoising (generation) process. The forward diffusion process incrementally introduces Gaussian noise to a raw EEG sample until it transforms into random noise. On the other hand, the reverse generation process serves as a denoising procedure, gradually removing noise from random noise until an EEG sample is generated. To model the denoising steps in the reverse process, a modified 1D-Unet was employed¹, facilitating the generation of EEG signals from noise. The quality of the generated EEG

¹The code is publicly available at <https://github.com/Typing784/denoising-diffusion-pytorch>.

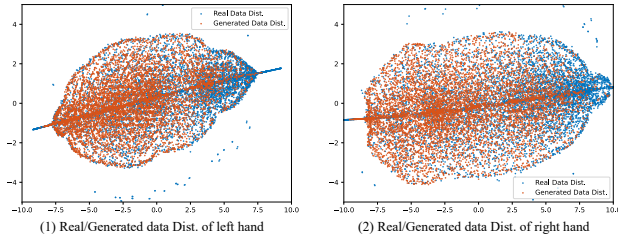


Fig. 2. Illustration of real and generated EEG data distributions of the 2-dimensional projection with t-SNE.

signals was assessed through both qualitative and quantitative analyses.

II. METHODOLOGY

A. Experimental Material and Paradigm

To facilitate the training of the generative model, an extensive collection of motor imagery data was initially collected. The study comprised a total of 64 subjects, and ethical approval was obtained from the Ethics Committee of Zhejiang University, Hangzhou, Zhejiang, China. Prior to engaging in the experiments, all participants provided their informed consent by signing consent forms. EEG signals were recorded through two systems: a 32-channel wireless iRecorderW32 system (Econtek, China) (23 subjects) and a 64-channel wireless NeuSenW64 system (Neural, China) (41 subjects). The channels were placed according to the international extended 10/20 system. As the data was acquired using different EEG caps, the common 32 channels shared by both caps were extracted for use. Among the 64 subjects, 49 comprised the training group for training data collection, while the remaining 15 formed the testing group. In the training group, each participant engaged in 3 to 6 runs of the motor imagery experiment, with each run consisting of 40 trials. In total, 7690 trials were employed for training the generative models. The testing group participants underwent 9 runs of the motor imagery experiment each.

B. EEG Signals Generation

Initially, we preprocessed the raw signals according to a standard preprocessing pipeline employing EEGLAB [10]. The raw EEG signals were re-referenced to the common average and subsequently filtered to the alpha-beta band within the frequency range of [8, 26] Hz. To mitigate variations arising from diverse subjects and recording devices, signal standardization was implemented for each trial [11]:

$$X'_i = M^{-1/2} X_i \quad (1)$$

where M represents the Euclidean mean derived from the covariance matrices of all trials of a specific subject. To facilitate the input of data into the model and the training of the model, the signals were scaled, constraining them to a range between -1 and 1. Two distinct DDPM models were

employed to autonomously model the distribution of EEG data for both left-hand and right-hand data sets.

DDPM, as a likelihood-based model, seeks to establish a model that assigns the maximum likelihood to observed data samples. When given observed EEG samples represented as $\mathbf{X}$ originating from an unknown distribution $P(\mathbf{X})$, DDPM aims to model the underlying data distribution $P(\mathbf{X})$ so that we can generate new samples approximately from $P(\mathbf{X})$. The diffusion model achieves this by defining two Markov chains: a forward diffusion process and a reverse denoising process. As shown in Figure 1, in the forward process, Gaussian noise is progressively added to the initial EEG samples $\mathbf{X}_0$ in T steps by (2) until it becomes identical to Gaussian noise:

$$\mathbf{X}_t = \sqrt{1 - \beta_t} \mathbf{X}_{t-1} + \sqrt{\beta_t} \epsilon_{t-1}, t = 1, \dots, T \quad (2)$$

where $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$ and $\{\beta_t \in (0, 1)\}_{t=1}^T$ is a variance schedule to control the variance of the Gaussian noise in each step. Because the resulting sequence $\{\mathbf{X}_0, \dots, \mathbf{X}_T\}$ forms a Markov chain, the conditional probability of $\mathbf{X}_t$ given $\mathbf{X}_{t-1}$ is also a Gaussian distribution:

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \mathcal{N}(\mathbf{X}_t; \sqrt{1 - \beta_t} \mathbf{X}_{t-1}, \beta_t \mathbf{I}) \quad (3)$$

In the reverse process, although $q(\mathbf{X}_{t-1} | \mathbf{X}_t, \mathbf{X}_0)$ is also a Gaussian distribution and can be directly calculated, when exclusively relying on the reverse process for sample generation, determining the condition $\mathbf{X}_0$ in advance is not feasible. Consequently, a U-net neural network, denoted as $p_\theta(\mathbf{X}_{t-1} | \mathbf{X}_t)$, is employed to model $q(\mathbf{X}_{t-1} | \mathbf{X}_t)$. Specifically, $p_\theta(\mathbf{X}_{t-1} | \mathbf{X}_t)$ is also a Gaussian distribution:

$$p_\theta(\mathbf{X}_{t-1} | \mathbf{X}_t) = \mathcal{N}(\mathbf{X}_{t-1}; \mu_\theta(\mathbf{X}_t, t), \Sigma_\theta(\mathbf{X}_t, t)) \quad (4)$$

At step t ($t \neq 0$ or T), the objective function is a KL divergence between $q(\mathbf{X}_{t-1} | \mathbf{X}_t)$ and the model $p_\theta(\mathbf{X}_{t-1} | \mathbf{X}_t)$. In practice, it can be formulas as the reconstruction loss:

$$L_t = \frac{1}{2} \left(\frac{\bar{\alpha}_{t-1}}{1 - \bar{\alpha}_{t-1}} - \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t} \right) \left[\left\| \hat{\mathbf{X}}_\theta(\mathbf{X}_t, t) - \mathbf{X}_0 \right\|_2^2 \right] \quad (5)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$. $\mathbf{X}_0$ is the real EEG signals, and $\hat{\mathbf{X}}_\theta$ is generated EEG signals using the trained network at step t . Once the network is trained, EEG samples can be progressively generated by denoising a random point drawn from $\mathcal{N}(0, \mathbf{I})$ through T steps:

$$\mathbf{X}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{X}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{X}_t, t) \right) + \sigma_t \mathbf{z}, \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}) \quad (6)$$

To synthesize EEG signals, DDPM is trained on the recorded signals from the subjects of the training group. To effectively capture both spatial and temporal information inherent in EEG signals, a modification was made to the original DDPM model. Specifically, the two-dimensional convolutional structure of the U-net was adjusted to a one-dimensional convolutional structure. During training, the variance of the noise was set to $\beta_{max} = 0.02$ and $\beta_{min} = 10^{-4}$, with a linear

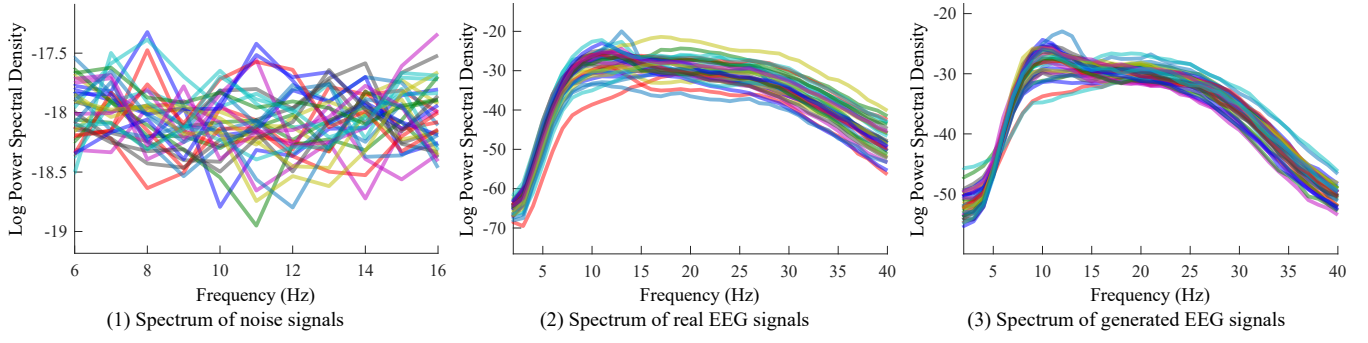


Fig. 3. Illustration of power spectral density of the noise, real, and generated EEG signals.

schedule, and the learning rate was set as 0.00008. The training process was performed over 10,000 iterations, utilizing a batch size of 800. The model was trained using 8 Nvidia RTX 4090 GPUs parallelly.

C. Classification Performance

We evaluated the quality of the generated samples from two perspectives. Firstly, by examining the separability of the generated samples, and secondly, by analyzing the distribution of discrimination information to ascertain its localization within the motor imagery-related cortex. The discriminative nature of the generated samples is represented by the correlation between the time-frequency decomposition results and their corresponding sample labels, quantified through the square of the correlation coefficient (R^2). Additionally, another performance evaluation was conducted, where the last 3 runs of the test group constituted the test set, and the generated data, combined with the first 6 runs, formed the training set. This evaluation employed the Common Spatial Pattern (CSP) [12] with linear discriminative analysis (LDA).

III. RESULTS

We performed both qualitative and quantitative analyses of the generated EEG signals. In the qualitative analysis, a comparison was made between the data distribution and power spectral density of the real EEG data and the generated EEG data. In the quantitative analysis, we assessed the separability of left-hand and right-hand tasks in the generated EEG data through classification and the distribution of discriminative information.

A. Quality of the Generated EEG

We conducted spectral analysis on randomly generated noise signals, real EEG signals, and generated EEG signals using the *spectopo()* function in EEGLAB. As shown in Figure 3, the energy in the noise signals is dispersed randomly across various frequency points. In contrast, the energy distribution of the generated signals closely resembles that of the real signals, both concentrating in the alpha-beta frequency band ([8, 26] Hz). Furthermore, we applied t-SNE [13] for dimension reduction on both the real and generated signals. Figure 2 illustrates the similarity in data distributions between the generated and real EEG signals in both the left-hand and right-hand.

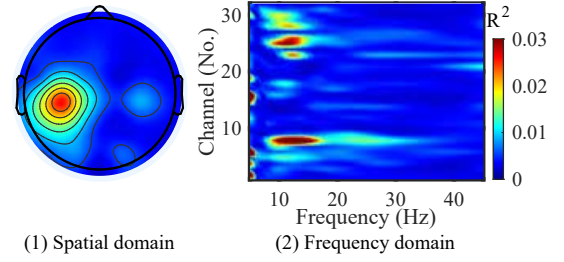


Fig. 4. Discriminative information distribution of generated EEG signals in spatial and frequency domain.

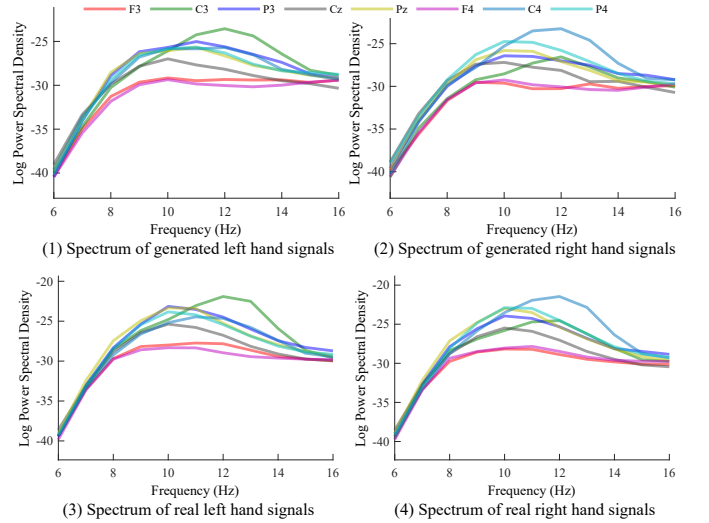


Fig. 5. Comparison of the event-related synchronization (ERS) in the generated and real EEG signals during the left-hand and right-hand motor imagery tasks.

B. Classes-related Information of Generated EEG

We conducted spectral analysis on the left-hand and right-hand classes of both the generated and real EEG signals. As depicted in Figure 5, a distinct event-related synchronization (ERS) phenomenon is evident. In the spectrum of left-handed signals, electrodes C3 and P3 in the left brain hemisphere manifest higher energy, while in the spectrum of right-handed signals, electrodes C4 and P4 in the left brain hemisphere showcase higher energy. Notably, the ERS phenomenon is lo-

TABLE I
NON-AUGMENTED AND AUGMENTED CLASSIFICATION PERFORMANCE OF
THE BCI-DEFICIENCY SUBJECTS.

Subjects	Non-Augmented (%)	Augmented (%)
1	58.33	60.83
2	69.17	72.50
3	60.00	67.50
4	64.17	71.67
5	68.33	63.33
Avg.	64.00 $\pm$ 4.84	67.17 $\pm$ 5.90

calized in the alpha band ([8, 14] Hz). Furthermore, employing ten-fold cross-validation based on the CSP-LDA algorithm for the generated EEG signals yielded an accuracy of $89.81 \pm 2.11\%$. Additionally, Figure 4 illustrates that the spatial and spectral distribution of discriminative information predominantly resides in the motor-sensory cortex and the alpha-beta frequency band. This observation aligns with the physiological basis of motor imagery [14].

C. Augmented Classification

To evaluate the effectiveness of generated EEG in addressing the BCI-deficiency issue, we initially trained the model using the data from the first six runs of each subject to select subjects exhibiting an accuracy rate below 70%. Subsequently, we integrated the generated EEG data with the initial six runs of the data for classifier training. The results are detailed in Table I, showcasing an average accuracy improvement of 3.17% (not statistically significant, $p = 0.1193$, paired t-test).

IV. DISCUSSION

This study explores a new method for generating EEG signals for MI-based BCI applications. The generated EEG signals and the real EEG signals have similar data distribution and power spectral density. This finding indicates that the reverse denoising process of the DDPM model effectively encodes information from real EEG signals. Notably, as shown in Figure 2, the real EEG signals contain some outliers [15], a drawback absent in the generated signal. This observation suggests potential advantages in the quality of generated EEG signals. Furthermore, the ERS shown in Figure 5 is a distinctive feature frequently observed in real EEG signals [16], which indicates that DDPM has encoded the motor imagery-related information.

The generated EEG signals exhibit a high classification performance between the two classes. Although the use of two distinct models for training and generation could potentially introduce prior information about classes, the results illustrated in Figure 4 indicate that the discriminative information primarily originates from the motor-sensory cortex and alpha-beta band, specifically associated with the motor imagery task. Therefore, the discriminative information is attributed more to the distinctions between left-hand and right-hand motor imagery tasks [17] rather than differences in the models themselves.

V. CONCLUSION

The Denoising Diffusion Probabilistic Model (DDPM) proves highly effective in generating EEG signals, highlighting its capacity to capture both spatial and spectral information. Additionally, the utilization of the generated EEG samples has the potential to improve the MI-based BCI classification performance for BCI-deficiency subjects. This study provides initial insights into the application of DDPM in EEG generation, offering foundational guidance for future research in this domain.

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